In JEE Advanced 2016, FIITJEE Students bag 36 in Top 100 AIR, 75 in Top 200 AIR, 183 in Top 500 AIR. 3541 Students from Long Term Classroom/ Integrated School Program & 4423 Students from All Programs have qualified in JEE Advanced, 2016

ALL INDIA TEST SERIES

FIITJEE

JEE (Advanced)-2017

FULL TEST – I

Paper 1

Time Allotted: 3 Hours **Maximum Marks: 243**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into three sections: **Section-A, Section-B & Section-C**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. **Section – A (01 – 06)** contains 6 multiple choice questions which have only one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

   Section – A (07 – 09) contains 3 multiple choice questions which have one or more than one correct answer. Each question carries **+4 marks** for correct answer. There is no negative marking.

   Section-A (10 – 14) contains 2 paragraphs. Based upon paragraph, 2 and 3 multiple choice questions have to be answered. Each question has only one correct answer and carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.
2. **Section – B (01 – 02)** contains 2 Matrix Match Type questions containing statements given in 2 columns. Statements in the first column have to be matched with statements in the second column. Each question carries **+8 marks** for all correct answer. For each correct row **+2 marks** will be awarded. There may be one or more than one correct matching. No marks will be given for any wrong matching in any question. There is no negative marking.
3. **Section-C (01 – 05)** contains 5 Numerical based questions with answers as numerical value from **0 to 9** and each question carries **+4 marks** for correct answer. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

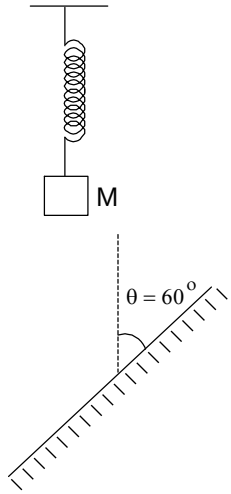
PART – I

SECTION – A

Straight Objective Type

This section contains 6 multiple choice questions numbered 1 to 6. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct

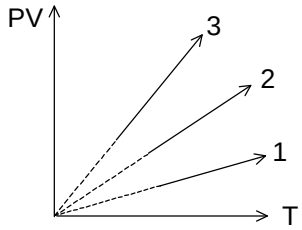
1. An insect of negligible mass is sitting on a block of mass M, tied with a spring of force constant K. The block performs simple harmonic motion with amplitude A infront of a plane mirror placed as shown. The maximum speed of insect relative to its image will be



(A) $A\sqrt{\frac{K}{M}}$ (B) $\frac{A\sqrt{3}}{2}\sqrt{\frac{K}{M}}$

(C) $A\sqrt{3}\sqrt{\frac{K}{M}}$ (D) $A\sqrt{\frac{M}{K}}$

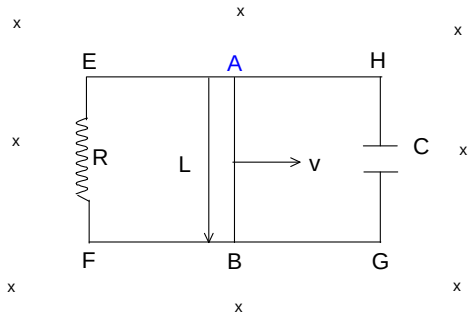
2. PV versus T graph of equal masses of H_2, He and CO_2 is shown in figure. Choose the correct alternative



(A) 3 corresponds to H_2, 2 to He and 1 to CO_2
(B) 1 corresponds to He, 2 to H_2 and 3 to CO_2
(C) 1 corresponds to He, 3 to H_2 and 2 to CO_2
(D) 1 corresponds to CO_2, 2 to H_2 and 3 to He

Space for Rough work

3. A conducting rod is moving with a constant velocity v over the parallel conducting rails which are connected at the ends through a resistor R and capacitor C as shown in the figure. Magnetic field B is into the plane. Consider the following statements.
(I) current in loop AEFBA is anticlockwise
(II) current in loop AHGBA is clockwise
(III) current through the capacitor is zero
(IV) energy stored in the capacitor is $\frac{1}{2}CB^2L^2V^2$
Which of the following option is correct
(A) statement (I) and (III) are only correct
(B) statement (II) and (IV) are only correct
(C) statement (I), (II) and (IV) are correct
(D) all the above

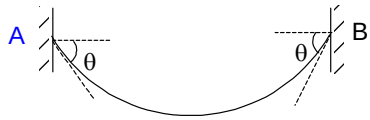


4. The activity of a radioactive sample goes down to about 6% in a time of 2 hours. The half life of sample in minutes is about
(A) 30 (B) 15 (C) 60 (D) 120

5. A particle of charge q and mass m starts moving from the origin under the action of an electric field $\vec{E} = E_0\hat{i}$ and magnetic field $\vec{B} = B_0\hat{k}$ its velocity at (x, 3, 0) is $\left(4\hat{i} + 3\hat{j}\right)$. The value of x is

(A) $\frac{36E_0B_0}{qm}$ (B) $\frac{25m}{2qE_0}$ (C) $\frac{10m}{qE_0}$ (D) $\frac{25E_0B_0}{m}$

6. A heavy string of mass m hangs between two fixed points A and B at the same level. The tangents to the string at A and B are at an angle θ with horizontal as shown in the figure. The tension of the string at its lowest point is



(A) $\frac{mg}{2\sin\theta}$ (B) $\frac{mg}{2\cos\theta}$
(C) $\frac{mg}{2\tan\theta}$ (D) $\frac{mg}{2\cot\theta}$

Space for Rough work

Multiple Correct Choice Type

This section contains 3 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct

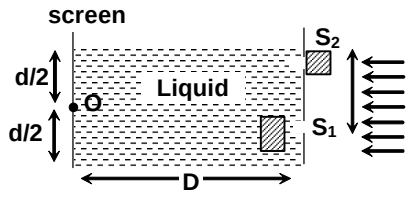
7. In a resonance tube experiment to determine speed of sound , air column in the pipe is made to resosnate with a given fork of frequency 480 Hz. At room temperature **T** using a resonance tube a student finds two resonance position 16.0 cm and 51.0 cm, length of the air column for the first and the second resonance respectively .It is known that speed of the sound at $273K$ is equal to 330 m/s.Then,

(A) percentage error in the measurement of speed of sound is $\frac{4}{7}\%$

(B) diameter of the tube is 5 cm
(C) temperature of the room **T** is nearly 283 K

(D) percentage error in the determination of **T** is $\frac{8}{7}\%$

8. A Young's double slit arrangement is immersed in a liquid of refractive index μ_1. The slit plane touches the liquid surface. A parallel beam of monochromatic light of wavelength 'λ' (with respect to air) is incident normally on slits. Choose the correct alternatives



(A) The fringe width is $D\lambda/\mu_1 d$.
(B) If the slit S_2 is covered by a transparent slab of refractive index μ_2 and thickness t, the new position of the central maxima will shift up by $\frac{(\mu_2 - 1).tD}{d}$
(C) The ratio of intensities at 'O' in the conditions of the option (A) and option the (B) is $1 : \cos^2\left[\frac{\pi}{3\lambda}(\mu_2 - 1)t\right]$
(D) If a slab of thickness t and refractive index μ_3 is introduced in front of S_1 while S_2 is also covered by a slab of thickness t and refractive index μ_2 as shown in figure, then for relation of $\mu_3 = \mu_1\mu_2$ the central maxima will be at O.

9. A net non-zero external force acts on a system of particles. At an instant t, the velocity and the acceleration of CM are found to be u_0 and a_0, it is possible that
(A) $u_0 = 0$, $a_0 = 0$ (B) $u_0 \neq 0$, $a_0 \neq 0$
(C) $u_0 \neq 0$, $a_0 = 0$ (D) $u_0 = 0$, $a_0 \neq 0$

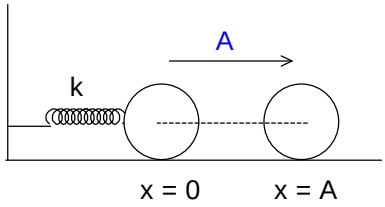
Space for Rough work

Comprehension Type

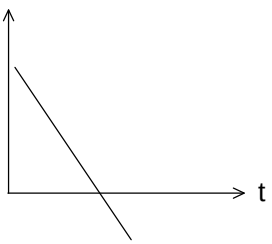
This section contains 2 paragraphs. Based upon one of paragraph 2 multiple choice questions and based on the other paragraphs 3 multiple choice questions have to be answered. Each of these questions has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct

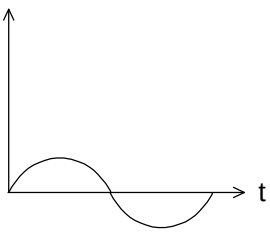
Paragraph for Question Nos. 10 and 11

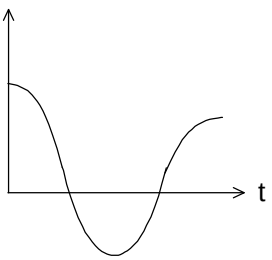
A disc of mass m and radius R is attached with a spring of force constant K at its centre as shown in figure. At x = 0 spring is unstretched. The disc is moved to x = A and then released. There is no slipping between disc and ground. Let f be the force of friction on the disc from the ground.

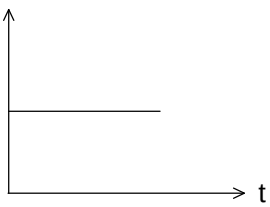


10. f versus t graph will be

(A)


(B)


(C)


(D)


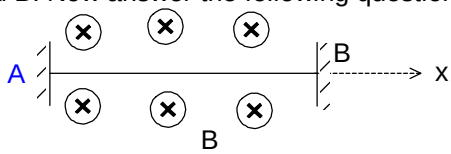
Space for Rough work

11. In the problem if k = 10 N/m, m = 2 kg, R = 1 m and A = 2 m. Find linear speed of the disc at mean position.

(A) $\sqrt{\frac{40}{3}}$ m/sec (B) $\sqrt{20}$ m/sec (C) $\sqrt{\frac{10}{3}}$ m/sec (D) $\sqrt{\frac{50}{3}}$ m/sec

Paragraph for Question Nos. 12 and 14

A standing wave $y = 2A\sin(kx)\cos(\omega t)$ set up in a wire AB fixed at both ends by two vertical wall contains a constant magnetic field B. Now answer the following questions.



12. The wire is found to vibrate in 3rd harmonic the maximum e.m.f induced is

(A) $\frac{4AB\omega}{k}$ (B) $\frac{3AB\omega}{k}$ (C) $\frac{2AB\omega}{k}$ (D) $\frac{AB\omega}{k}$

13. In above question the time when the e.m.f becomes maximum for first time is

(A) $\frac{2\pi}{\omega}$ (B) $\frac{\pi}{\omega}$ (C) $\frac{\pi}{2\omega}$ (D) $\frac{\pi}{4\omega}$

14. In which of following modes the e.m.f induced in AB is always zero.

(A) Fundamental mode (B) second harmonic (C) second overtone (D) fourth overtone

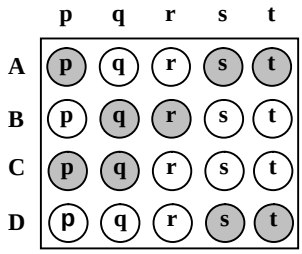
Space for Rough work

SECTION – B

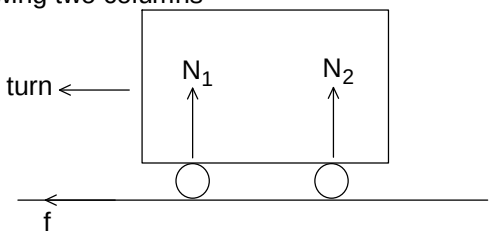
(Matrix-Match Type)

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:



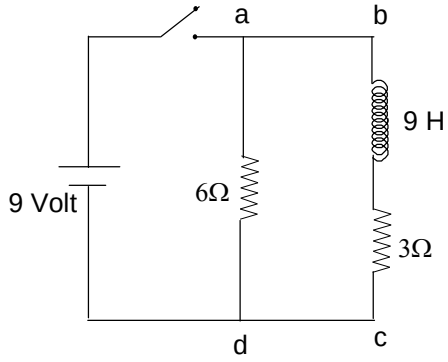
1. A car is taking a turn on a rough horizontal road without slipping as shown in figure. Let F is centripetal force, f the force of friction, N_1 and N_2 are two normal reactions. As the speed of car is increased match the following two columns



Column – I		Column – II	
(A)	N_1	(p)	Will increase
(B)	N_2	(q)	Will decrease
(C)	F/f	(r)	Will remains unchanged
(D)	F	(s)	Cannot say anything

Space for Rough work

2. In the circuit shown in figure, switch remains closed for long time. It is opened at time t = 0. Match the following columns at t = ln2 sec



Column – I		Column – II	
(A)	Potential difference across inductor	(p)	9 V
(B)	Potential difference across 3Ω resistor	(q)	4.5 V
(C)	Potential difference across 6Ω resistor	(r)	6 V
(D)	Potential difference between points b and c	(s)	None of these

Space for Rough work

SECTION – C

(One Integer Value Correct Type)

This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. A frictionless tunnel is dug inside the uniform spheres as shown in the figure-1 and figure-2. A particle of mass m is to be projected from point P in both sphere so that it can escape from the influence of sphere. Let v_1 is the velocity given to the particle in figure-1 and v_2 is the velocity given to the particle in figure-2. Find the ratio of v_1/v_2. (Given: R is radius of sphere and M is mass of sphere.)

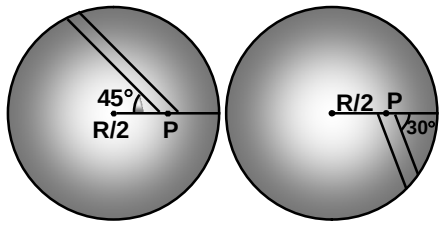
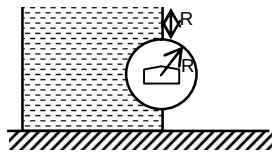


Figure-1 Figure-2

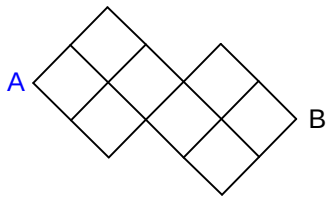
2. An α particle and proton are accelerated from rest by a potential difference of 100 V. After this their de-Broglie wavelengths are λ_α and λ_p respectively. The ratio $\frac{\lambda_p}{\lambda_\alpha}$ to the nearest integer is

3. A cylinder of radius R is kept embedded along the wall of a dam as shown. Take density of water as ρ and length of cylinder as L. The vertical force exerted by water on the cylinder is $\frac{\rho g L R^2 \pi}{X}$. Find the value of X.



4. A sphere of mass M and radius r rolls without slipping on a rough concave surface of large radius R. It makes small oscillation about lowest point. Find the value of k if time period is $\sqrt{\frac{k_1(R-r)}{5g}}$

5. Each branch in the following circuit has a resistance r = 1Ω. The equivalent resistance of the circuit between the points A and B is



Space for Rough work

Chemistry

PART – II

SECTION – A

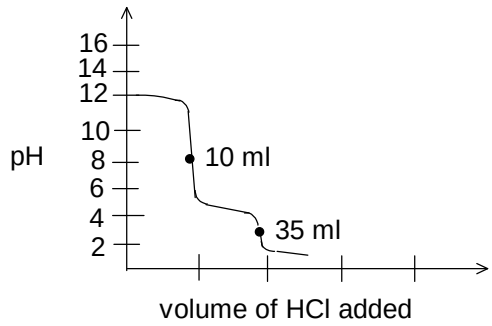
Straight Objective Type

This section contains 6 multiple choice questions numbered 1 to 6. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct

1. The equilibrium p-xyloquinone + methylene white $\rightleftharpoons$ p-xylohydroquinone + methylene blue may be conveniently by observing the difference in color methylene white and methylene blue. One mill mole of methylene blue was added to 1 L solution that was 0.24 M in p-xylohydroquinone and 0.012 M is p-xyloquinone. It was then found that 4% of the added methylene blue was reduced to methylene white. What is the equilibrium constant of the above reaction?
(A) 48 (B) 4.8 (C) 480 (D) 0.48

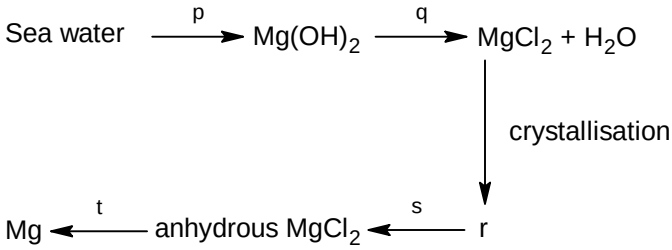
2. Sea water is found to contain 5.85% NaCl and 9.50% $MgCl_2$ by weight of solution. Calculate its normal boiling point assuming 80% ionization for NaCl and 50% ionization of $MgCl_2$. $\left[K_{b(H_2O)} = 0.51\,kg\,mol^{-1}K\right]$
(A) 102.29°C (B) 102.92 °C
(C) 101.59 °C (D) 101 °C

3. 0.50 gm mixture containing $NaHCO_3$, Na_2CO_3 and KCl was dissolved in 50 ml water and titrated with 0.15 M HCl resulting in the following titration curve. Determine weight % of $NaHCO_3$, Na_2CO_3 and KCl in the mixture respectively.
(A) 37.8, 31.8, 30.4
(B) 31.8, 37.8, 30.4
(C) 30.4, 37.8, 31.8
(D) 31.8, 30.4, 37.8



Space for Rough work

4. Extraction of Mg from sea water



Which of the following option describes the correct reactant, products and reaction conditions?

	p	q	r	s	t
(A)	Milk of magnesia	HCl (aq)	$MgCl_2.2H_2O$	Heating in limited supply of air	Reduction by coke
(B)	Lime water	HCl (aq)	$MgCl_2$	Heating in absence of air	Electrolytic reduction in molten state
(C)	Milk of lime	HCl (aq)	$MgCl_2.6H_2O$	Heating in presence of dry HCl	Electrolytic reduction in molten state
(D)	Milk of lime	HCl (aq)	$MgCl_2.6H_2O$	Heating in the presence of dry HCl	Electrolysis of aqueous solution

5. Which of the following statement is correct?
(A) geometrical isomerism is not observed in complexes of co-ordination number 4 having tetrahedral geometry
(B) square planar complexes generally do not show geometrical isomerism
(C) the square planar complex of general formula Ma_3b or Mab_3 exhibits cis-trans isomerism. (where M is central metal and a and b are monodentate ligands)
(D) the platinum glycinato complex $[Pt(Gly)_2]$ doesn't show geometrical isomerism.

6. Which of the following statement is correct?
(A) in an anti-fluorite structure anions form FCC lattice and cations occupy all tetrahedral voids.
(B) if the radius of cations and anions are 0.214 $\mathring{A}$ & 0.95 $\mathring{A}$ then co-ordinate number of cation in the crystal is 4
(C) an atom/ion is transferred from a lattice site to an interstitial position in Schottky defect
(D) density of crystal always increases due to substitutional impurity defect.

Space for Rough work

Multiple Correct Choice Type

This section contains 3 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct

7. Pick out the correct statements among the following from inspection of standard reduction potentials

$Cl_2(aq) + 2e \rightleftharpoons 2Cl^-(aq)$ $E^o_{Cl_2/Cl^-} = +1.36$ Volt

$Br_2(aq) + 2e \rightleftharpoons 2Br^-(aq)$ $E^o_{Br_2/Br^-} = +1.09$ Volt

$I_2(aq) + 2e \rightleftharpoons 2I^-(aq)$ $E^o_{I_2/I^-} = +0.54$ Volt

$S_2O_8^{2-}(aq) + 2e^- \rightleftharpoons 2SO_4^{2-}(aq)$ $E^o_{S_2O_8^{2-}/SO_4^{2-}} = +2.00$ Volt

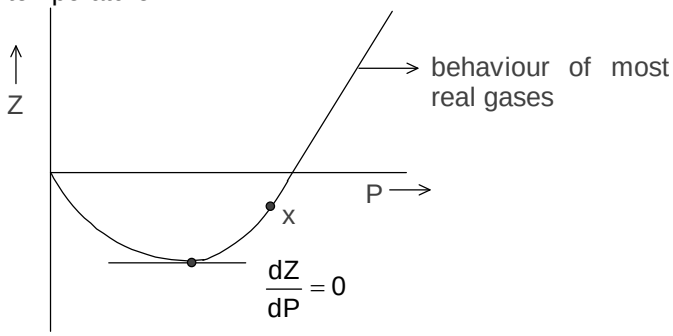
(A) Cl_2 can oxidise SO_4^{2-} from solution

(B) Cl_2 can oxidise Br^- and I^- from aqueous solution

(C) $S_2O_8^{2-}$ can oxidise Cl^-, Br^- and I^- from their aqueous solution

(D) $S_2O_8^{2-}$ is added slowly, Br^- can be reduced in the presence of Cl^-

8. Which of the following statement(s) are true about Z Vs P graph for a real gas at a particular temperature



(A) $Z \to 1$ as $P \to 0$ for most real gases

(B) $\frac{dz}{dp}$ is –ve as $P \to 0$ for most real gases

(C) $\frac{dz}{dp} \to 0$ at a pressure where replusive and attractive forces are comparable

(D) at point x, intermolecular attractive forces dominating over repulsive forces

Space for Rough work

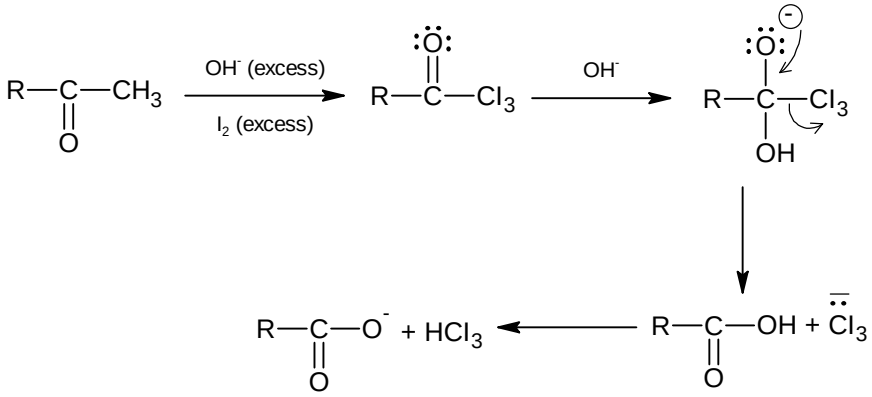
9. Saturated solution of SO_2 is heated at 150^oC in a closed container. The product obtained is treated with $BaCl_2$ solution. What is/are the observation(s)?
(A) no ppt. (B) white turbidity (C) evolution of SO_2 (D) white ppt.

Comprehension Type

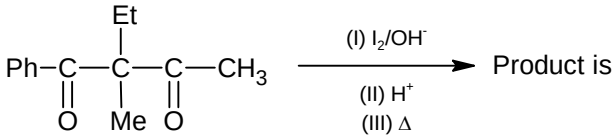
This section contains 2 paragraphs. Based upon one of paragraph 2 multiple choice questions and based on the other paragraphs 3 multiple choice questions have to be answered. Each of these questions has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct

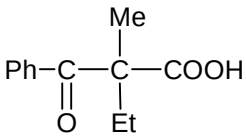
Paragraph for Question Nos. 10 and 11

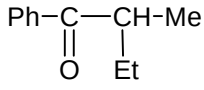
In the presence of excess base and excess halogen a methyl ketone is converted first into a trihalo substituted ketone and then into a carboxylic acid. After the trihalo substituted ketone is formed hydroxide ion attacks the carboxyl carbon because the trihalomethyl ion is the group more easily expelled from the tetrahedral intermediate. The conversion of a methyl ketone to a carboxylic acid is called a haloform reaction because one of the product is haloform $CHCl_3$ or CHI_3 or $CHBr_3$.

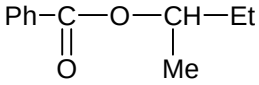


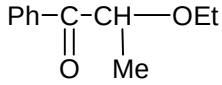
10.



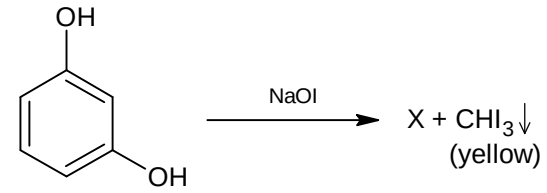
(A) 

(B) 

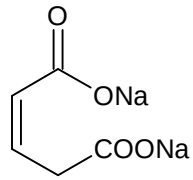
(C) 

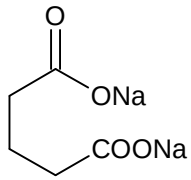
(D) 

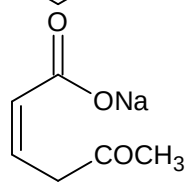
Space for Rough work

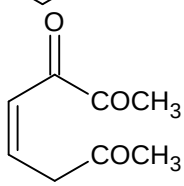
11. 

X is

(A) 

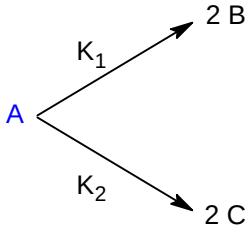
(B) 

(C) 

(D) 

Paragraph for Question Nos. 12 to 14

For a hypothetical elementary reaction



Where $\frac{K_1}{K_2} = \frac{1}{2}$, initially only 2 moles of A are present

12. The total number of moles of A, B & C at the end of 50% reaction are
(A) 2 (B) 3 (C) 4 (D) 5

Space for Rough work

13. Number of moles of B at the end of 75% reaction is
(A) 2 (B) 1 (C) 0.50 (D) 1.5

14. The ratio of number of moles of B to C at the end of 100% reaction is
(A) 2 : 1 (B) 1 : 2 (C) 1 : 3 (D) 2 : 3

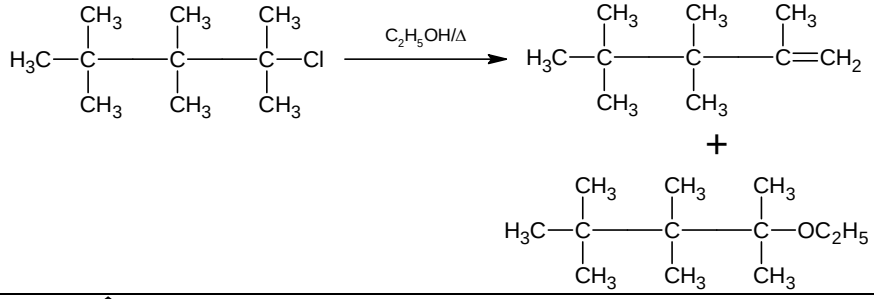
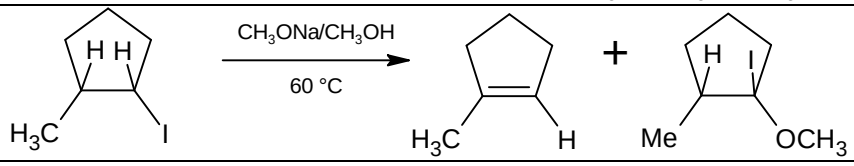
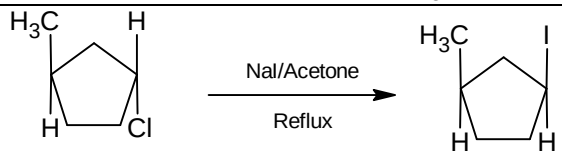
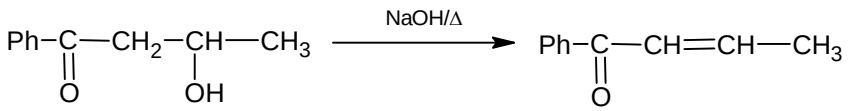
SECTION – B

(Matrix-Match Type)

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:
If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:

	p	q	r	s	t
A	●	○	○	●	●
B	○	●	●	○	○
C	●	●	○	○	○
D	○	○	○	●	●

1. Match the following.

Column – I		Column–II	
(A)		(p)	SN_1
(B)		(q)	E_1CB
(C)		(r)	E_1
(D)		(s)	E_2
		(t)	SN_2

Space for Rough work

2. Match the Following

Column – I		Column - II	
(A)	Reversible adiabatic compression	(p)	$\Delta S_{system} > 0$
(B)	Reversible vaporization	(q)	$\Delta S_{system} < 0$
(C)	Adiabatic free expansion of ideal gas in vaccum	(r)	$\Delta S_{surrounding} < 0$
(D)	Dissociation of $CaCO_3(s) \longrightarrow CaO(s) + CO_2(g)$	(s)	$\Delta S_{surrounding} = 0$
		(t)	$\Delta S_{system} = 0$

SECTION – C

(One Integer Value Correct Type)

This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. For 1s orbital of hydrogen atom radial wave function is given as

$$R_{(r)} = \frac{1}{\sqrt{\pi}}\left(\frac{1}{a_0}\right)^{\frac{3}{2}} e^{-\frac{r}{a_0}}$$ (where $a_0 = 0.529\,\mathring{A}$)

The ratio of radial probability density of finding electron at $r = a_0$ to the radial probability density of finding electron at nucleus is given as $x.e^{-y}$. Calculate the value of (x + y).

2. 4 g isolated Ca atom in gaseous state is provided a total energy of 50 kJ. The molar ratio of Ca^{+}/Ca^{+2} ions produced is x/y then x + y = …….
IE_1 of Ca = 400 kJ/mole and IE_2 of Ca = 700 kJ/mole

3. The oxidation state of metal in Fehling solution, on the reaction with benzaldehyde changes by ……. units.

4. A decapeptide (Mol. wt. 796) on complete hydrolysis gives glycine (mol. wt. 75), alanine and phenylalanine. Glycine contributes 47% to the total weight of the hydrolysed products. The number of glycine units present in the decapeptide is

5. PDI (poly dispersity index) for natural polymers is generally close to …..

Space for Rough work

Mathematics

PART – III

SECTION – A

Straight Objective Type

This section contains 6 multiple choice questions numbered 1 to 6. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct

1. P_1 is a parabola $y^2 = 4ax$, and P_2 is a parabola given by $y^2 = 4b(x + a)$. P_2 passes through A and B which are points of intersection of P_1 and line $x = a$. If T is the point of intersection of tangents to P_1 at A and B, and R is the point of intersection of normal at A and B to P_2 the area of quadrilateral TARB is

   (A) $\sqrt{5}\,a^2$ (B) $2\sqrt{5}\,a^2$ (C) $3a^2$ (D) $6a^2$

2. The solution of the differential equation $\frac{dy}{dx} = -\frac{y\sin(x+y) + xy\cos(x+y) - x^2y^2}{x\sin(x+y) + xy\cos(x+y) - x^2y^2}$.

   (A) $xy\sin(x+y) = c_1$ (B) $xy\sin(x+y+c_1) = c_2$; c_1, c_2 are related

   (C) $x^2y^2\sin(x+y) = c_1$ (D) $x^2y^2\sin(x+y+c_1) = c_2$; c_1, c_2 are related

3. The number of proper divisors of $K = \sum_{i\le}\sum_{j\le}\sum_{n\le 10}\left({}^{10}C_n\,{}^{n}C_j\,{}^{j}C_i\right)$ is

   (A) 19 (B) 20 (C) 21 (D) none of these

4. In a matrix $A_{4\times 4}$, an ant starts moving from a_{11} in such a way that it can move from a_{ij} to either $a_{(i+1)j}$ or $a_{i(j+1)}$ in one step. If it reaches a_{44}, the probability that it passes through a_{23} is

   (A) $\frac{18}{35}$ (B) $\frac{1}{4}$ (C) $\frac{9}{20}$ (D) none of these

5. If $K \in R$, $10^K.\left(x^{\log_{10}^2 x - K\log_{10} x - 1}\right) = 1$. The maximum number of solutions of the equation for 'x' is

   (A) 0 (B) 1 (C) 3 (D) 4

Space for Rough work

6. The number of solution in x of the equation
$\sqrt{\sqrt{3}f(x)-f'(x)-2}+\sqrt{\cot 3x+\text{cosec}^2 x-4}=\sin\left(\frac{3x}{2}\right)+\frac{\sqrt{2}}{2}$ if $a\in[0,6\pi]$ and f(x) satisfies the differential equation $f''(x)+f(x)=0$ and $(f'(x))^2+(f(x))^2=1$, and f(0) = 1 is
(A) 1 (B) 3 (C) 4 (D) 0

Multiple Correct Choice Type

This section contains 3 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct

7. A square is inscribed in a circle. If p_1 is the probability that a randomly chosen point of the circle lies within the square and p_2 is the probability that the point lies outside the square, then
(A) $p_1 < p_2$ (B) $p_1 = p_2$ (C) $p_1 > p_2$ (D) $p_1^2 - p_2^2 < \frac{1}{3}$

8. If $I_{m,n}=\int \sec^{2n} x\tan^m x\,dx$, $n\geq 1$ and $m\geq 2$, m, $n\in N$, then
(A) $I_{m,n}=\frac{\sec^{2n-2}x\tan^{m+1}x}{m+1}-\frac{(n-1)}{m+1}I_{m+2,\,n-1}$ (B) $I_{m,n}=\frac{\sec^{2n}x\tan^{m-1}x}{2n}-\frac{m-1}{2n}I_{m-2,\,n+1}$
(C) $\frac{m-1}{n}I_{m+2,n-1}+\frac{1}{m+1}=\frac{1}{n}+\frac{2(n-1)}{m+1}I_{m-2,n+1}$ (D) $\frac{m-1}{2n}I_{m+2,n-1}+\frac{1}{m+1}=\frac{1}{2n}+\frac{2(n-1)}{m+1}I_{m-2,n+1}$

9. If $\vec{a},\vec{b}$ and $\vec{c}$ are three unit vectors and $\vec{d},\vec{e}$ and $\vec{f}$ are three linearly independent unit vectors such that $(\vec{a}.\vec{b})\vec{d}+(\vec{b}.\vec{c})\vec{e}+(\vec{c}.\vec{a})\vec{f}=0$, then
(A) $k\vec{a}$, $\ell\vec{b}$ and $m\vec{c}$ are linearly independent (B) $[\vec{d}\,\vec{e}\,\vec{f}]=\pm 1$
(C) $[\vec{a}\,\vec{b}\,\vec{c}]=\pm 1$ (D) $[\vec{a}+(\vec{b}\times\vec{c})].(\vec{b}+\vec{c})=0$

Space for Rough work

Comprehension Type

This section contains 2 paragraphs. Based upon one of paragraph 2 multiple choice questions and based on the other paragraphs 3 multiple choice questions have to be answered. Each of these questions has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct

Paragraph for Question Nos. 10 and 11

A circle C_1 touches the circle $x^2 + y^2 - a^2 = 0$ at A internally and the parabolas $y^4 - 16x^2 = 0$ at B and C externally.

10. If ΔABC formed is an equilateral then AB =

(A) $\frac{2}{\sqrt{3}}$ (B) $\frac{1}{3}$ (C) $\frac{2}{3}$ (D) $\frac{1}{\sqrt{3}}$

11. Which of the below is true?

(A) $|a| = \sqrt{3} - \frac{1}{\sqrt{3}}$ (B) radius of $C_1 = \frac{2}{\sqrt{3}}$ (C) $|a| = 2\sqrt{3}$ (D) radius of $C_1 = \frac{2}{3\sqrt{3}}$

Paragraph for Question Nos. 12 and 14

Let $\alpha_1, \alpha_2, ..., \alpha_n$ be the roots of the equation $f(x) = x^n + p_1 x^{n-1} + ... + p_{n-1}x + p_n = 0$

$\frac{f(x)}{x - \alpha_i} = x^{n-1} + (\alpha_i + p_1)x^{n-2} + (\alpha_i^2 + \alpha_i p_1 + p_2)x^{n-3} + ...$ Then $\sum_{i=1}^{n} \frac{f(x)}{x - \alpha_i} = f'(x)$ is an identity where

$f'(x) = \frac{df(x)}{dx}$

12. If $f(x) = x^5 + px^3 + qx + t = 0$, $\sum_{i=1}^{5} \alpha_i^3$

(A) $-p^3 + 3q$ (B) 1 (C) 0 (D) $-p^3 + 7q$

Space for Rough work

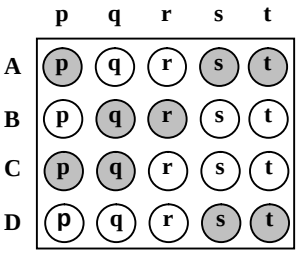
13. If $f(x) = x^3 - 2x^2 + x - 1$ and α_i, i = 1, 2, 3 are three roots of f(x) = 0 then $\sum \alpha_i^3$ is equal to
(A) 3 (B) 4 (C) 5 (D) 6

14. If $f(x) = x^3 - 2x^2 + x - 1 = 0$, $\sum \alpha_i^4 =$
(A) 10 (B) 15 (C) 20 (D) 25

SECTION – B

(Matrix-Match Type)

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:
If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:



1. Match the following:

Column – I		Column – II	
(A)	The last three digits of $3^{50} - 7^{20}$ contains	(p)	1
(B)	Divisors of the number of solution of the equation $\lvert\tan x\rvert = \{x\}$, $x \in [0,4]$ is/are ({x} is fractional part of x)	(q)	2
(C)	Divisors of the number of points of discontinuity of $f(x) = \lim_{n\to\infty} \frac{\{(x-1)(x-3)(x-5)\}^{2n} - 3^{2n}}{\{(x-1)(x-3)(x-5)\}^{2n} + 3^{2n}}$	(r)	4
(D)	$f(x) = \begin{cases} Kx, & x < 0 \\ x + \{x\}, & x \in [0,4] \\ \frac{K+8}{2} + (x-4)^2, & x > 4 \end{cases}$ has local minima at x = 0 and x = 4, then possible value of |K|, which are integer	(s)	6
		(t)	8

Space for Rough work

2. Match the following:

Column – I		Column – II	
(A)	$7\{x\}^4+9\{x\}^3\{-x\}-5\{x\}^2+9\{x\}\{-x\}-2=0$, {x} is factional part of x, then if $m \in z^+$, 2mx may be equal to	(p)	0
(B)	If $2y^2 + 2xy - 5x = 0$ has non-negative integral solutions in x and y then y may be equal to	(q)	2
(C)	A = {1, 3, 5, 7} and B = {2, 4, 6, 8} number of one-one function f: A $\to$ B such that $f(i)-i \geq 2\ \forall\ i \in A$	(r)	3
(D)	If $f(x)=\begin{cases} x+\sin x, & x\in(0,\pi) \\ 0, & x=\pi \\ \pi+1-\dfrac{1}{1+(x-\pi)^2}, & x\in(\pi,\infty) \end{cases}$ then the value of x for which $f(x) = f^{-1}(x)$	(s)	4
		(t)	5

SECTION – C
(One Integer Value Correct Type)

This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. If the number of ordered triplets (a, b, c) such that L.C.M (a, b) = 1000, L.C.M (b, c) = 2000 and L.C.M (c, a) = 2000 be N then sum of digits of N is/are _____

2. If $\arg\left(z^{2n}+\left(n^2+1\right)z^n+1\right)=kn\arg(z)$, |z| = 1, $n \in N$ then k equal …

3. If $f(x)=\begin{cases} e^x+e^{-x}, & x<0 \\ \log_{10}(100-x), & x\in[0,100) \end{cases}$ then the number of solution of the equation $f(x)+\left(e-\dfrac{1}{e}\right)x=\dfrac{2}{e}$

4. The equation of a plane which passes through the line of intersection of the planes x + y + z – 1 = 0 and x – y + z – 3 = 0 and which is at maximum distance from the point (3, 4, 5) is ax + by + cz – d = 0, the value of $\left(\dfrac{a+b-c}{d}\right)$ is

5. The quadratic equation $2x^2 - 5\pi x + p = 0$ has roots equal to $2\sin^{-1}\alpha$ and $3\sin^{-1}\beta$, then $\left[\dfrac{p}{\pi}\right]$ equals, [.] denotes greatest integer function.

Space for Rough work